

Calculus

Inequalities, Derivatives, and Value Theorems

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Chebyshev's Integral Inequality

Theorem

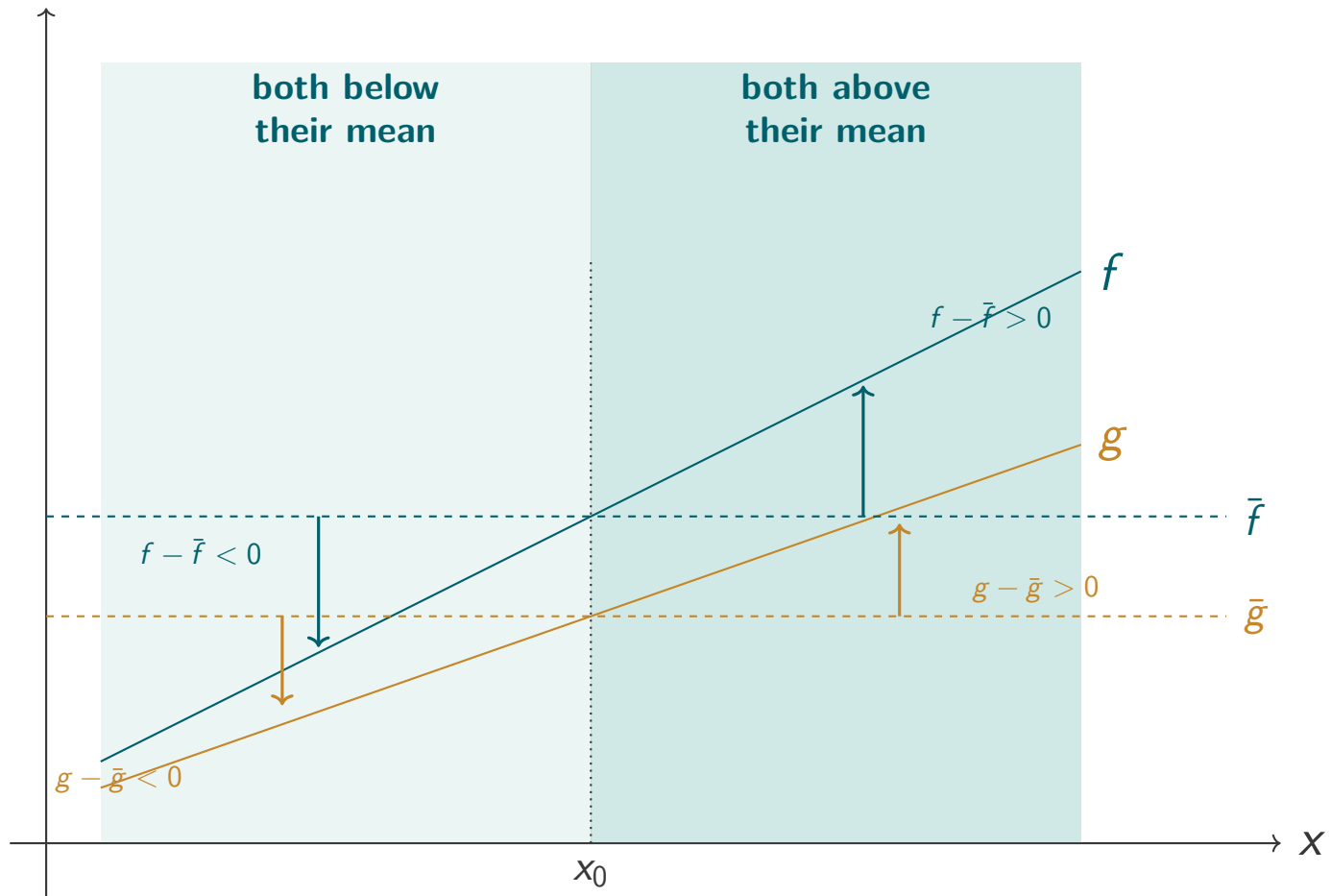
If f and g are both increasing (or both decreasing) on $[a, b]$, then

$$\frac{1}{b-a} \int_a^b f(x)g(x) dx \geq \left(\frac{1}{b-a} \int_a^b f dx \right) \left(\frac{1}{b-a} \int_a^b g dx \right).$$

If one is increasing and the other decreasing, the inequality reverses.

In words: when two functions rise together, the average of their product exceeds the product of their averages.

Geometric Picture: Chebyshev



The deviations $f(x) - \bar{f}$ and $g(x) - \bar{g}$ share a sign at every x , so $(f - \bar{f})(g - \bar{g}) \geq 0$. Averaging gives $\overline{fg} - \bar{f}\bar{g} \geq 0$, which is Chebyshev.

Application — Statement

Application

Prove that for every $n \in \mathbb{N}$,

$$\int_0^1 x^n e^x dx \geq \frac{e-1}{n+1}.$$

Both factors of the integrand are increasing on $[0, 1]$, so Chebyshev separates the product into a product of two elementary integrals — exactly the setting in which the inequality applies.

Application — Proof

Proof

Let $f(x) = x^n$ and $g(x) = e^x$. Both are strictly increasing on $[0, 1]$, so Chebyshev's inequality with $b - a = 1$ gives

$$\int_0^1 x^n e^x dx \geq \left(\int_0^1 x^n dx \right) \left(\int_0^1 e^x dx \right).$$

Evaluating each factor,

$$\int_0^1 x^n dx = \frac{1}{n+1}, \quad \int_0^1 e^x dx = e - 1,$$

hence

$$\int_0^1 x^n e^x dx \geq \frac{e-1}{n+1}. \quad \square$$

An Innocent-Looking Question

Question

Can the floor function $f(x) = \lfloor x \rfloor$ be the derivative of some differentiable function on $\mathbb{R}$?

Recall $\lfloor x \rfloor$ is constant on each interval $[n, n + 1)$ and jumps by 1 at every integer. Its antiderivative on each piece is a linear function with slope n , and we can certainly piece these together into a continuous function. Why might that not be a derivative?

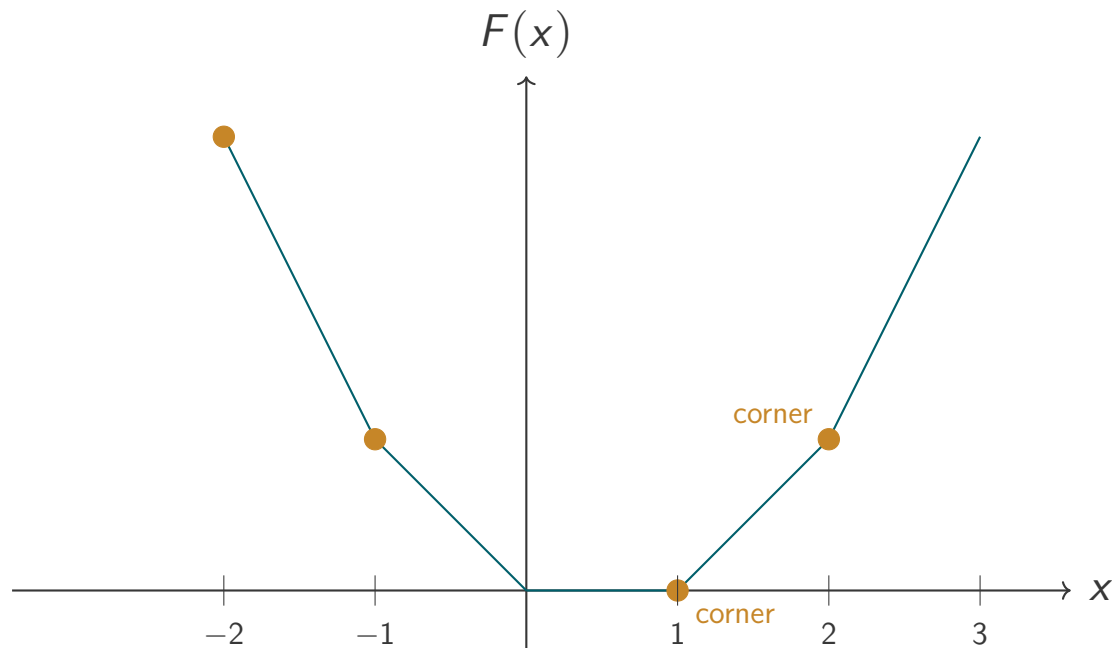
Pause and think about what the antiderivative looks like at the integer joins.

The Answer: No

The floor function cannot be a derivative on $\mathbb{R}$. The natural antiderivative

$$F(x) = \int_0^x \lfloor t \rfloor dt$$

is continuous, but at each integer n the left slope is $n - 1$ and the right slope is n . So F has a *corner* (not a tangent) at every integer; $F'(n)$ does not exist.



F is continuous but sharp-edged at each integer — so F is not differentiable there.

Why? — Darboux's Theorem

Darboux's Theorem

If F is differentiable on an interval I , then F' has the *intermediate value property* on I : for any $a, b \in I$ and any v strictly between $F'(a)$ and $F'(b)$, there exists $c \in (a, b)$ with $F'(c) = v$.

Consequence: **a derivative cannot have any jump discontinuity.** If F' jumped from L to $R \neq L$ at some point, it would skip every value strictly between L and R in any neighbourhood — violating the intermediate value property.

Since $\lfloor x \rfloor$ jumps by 1 at every integer, it has jump discontinuities, so it cannot be the derivative of any function on $\mathbb{R}$.

Types of Discontinuity

A function f can fail to be continuous at $x = a$ in three qualitatively different ways:

1. Removable. Both one-sided limits exist and are equal, but $f(a)$ is missing or wrong:

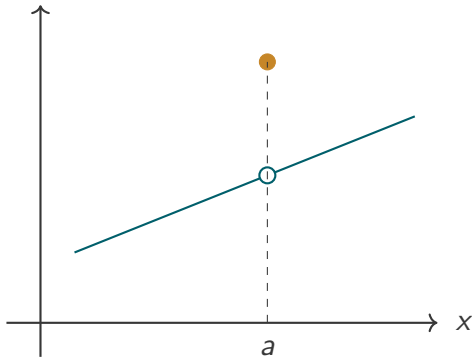
$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) \neq f(a) \quad (\text{or } f(a) \text{ undefined}).$$

2. Jump. Both one-sided limits exist but are unequal:

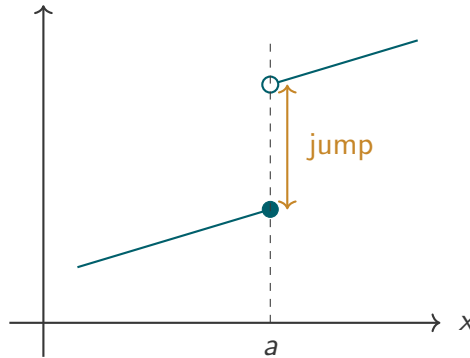
$$\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x).$$

3. Essential (infinite or oscillatory). At least one one-sided limit fails to exist (it diverges to $\pm\infty$, or oscillates without limit).

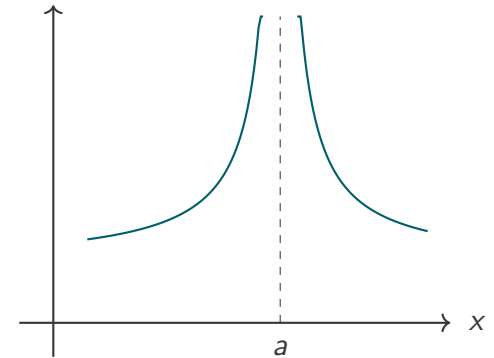
Discontinuity Diagrams



Removable
limits agree,
value disagrees



Jump
both limits exist,
unequal



Essential
a one-sided limit
diverges or oscillates

Examples. Removable: $f(x) = \frac{\sin x}{x}$ at $x = 0$ (define $f(0) = 1$). Jump: $\text{sgn}(x)$ at $x = 0$. Essential (infinite): $1/x$ at $x = 0$. Essential (oscillatory): $\sin(1/x)$ at $x = 0$.

MCQ — Which Can Be a Derivative?

Question

Which one of the following functions can be the derivative of some differentiable function on all of $\mathbb{R}$?

(A) $f(x) = |x|$

(B) $f(x) = \operatorname{sgn}(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ +1 & x > 0 \end{cases}$

(C) $f(x) = \{x\} = x - \lfloor x \rfloor$ (fractional part)

(D) $f(x) = \lfloor x \rfloor$ (floor function)

Hint: by Darboux's theorem, a derivative on $\mathbb{R}$ cannot have any jump discontinuity. Three of these functions jump; one does not.

MCQ — Answer

Answer: (A), $f(x) = |x|$.

Why $|x|$ works, the others don't

$|x|$ is continuous on $\mathbb{R}$ with no jumps, and indeed it *is* a derivative: take

$$F(x) = \frac{1}{2}x|x| = \begin{cases} \frac{1}{2}x^2, & x \geq 0, \\ -\frac{1}{2}x^2, & x < 0. \end{cases}$$

Then F is differentiable everywhere on $\mathbb{R}$ (including $F'(0) = 0$), and $F'(x) = |x|$. The other three all have jump discontinuities, so by Darboux's theorem none can be a derivative on any interval containing the jump:

- ▶ $\operatorname{sgn}(x)$ jumps from -1 to $+1$ at $x = 0$.
- ▶ $\{x\}$ jumps from 1 to 0 at every integer.
- ▶ $\lfloor x \rfloor$ jumps by 1 at every integer.

Problem 1 — Statement

Problem 1

Let A, B be given 3×3 real matrices. Show that

$$\det(A \sin x + B \cos x) = 0$$

for some real x .

Problem 1 — Proof

Proof

Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \det(A \sin x + B \cos x).$$

Each entry of $A \sin x + B \cos x$ is a continuous function of x , and the determinant is a polynomial in those entries, so f is continuous on $\mathbb{R}$.

Evaluating at two convenient points,

$$f(0) = \det(B), \quad f(\pi) = \det(-B) = (-1)^3 \det(B) = -\det(B).$$

Hence $f(0)$ and $f(\pi)$ have opposite signs (or both vanish). If $\det(B) = 0$, take $x = 0$. Otherwise $f(0)$ and $f(\pi)$ have strictly opposite signs, and by the intermediate value theorem there exists $x \in (0, \pi)$ with $f(x) = 0$. $\square$

Problem 2 — Statement

Problem 2

Let A be an $n \times n$ real matrix with n odd, and suppose $\det(A) > 0$. Prove that there exists a strictly negative real number c such that

$$\det(I + cA) = 0,$$

i.e. $I + cA$ is singular.

Problem 2 — Proof

Proof

Define $f(x) = \det(I + xA)$. As a polynomial in the entries of $I + xA$, f is a polynomial in x , hence continuous on $\mathbb{R}$. At $x = 0$, $f(0) = \det(I) = 1 > 0$. For $x \neq 0$, factor x out of each row of $I + xA$:

$$f(x) = \det(xA + I) = x^n \det\left(A + \frac{1}{x}I\right).$$

As $|x| \rightarrow \infty$, $\frac{1}{x}I \rightarrow 0$, so by continuity of the determinant

$$\det\left(A + \frac{1}{x}I\right) \longrightarrow \det(A) > 0.$$

Since n is odd, $x^n \rightarrow -\infty$ as $x \rightarrow -\infty$, hence $f(x) \rightarrow -\infty$ as $x \rightarrow -\infty$.

In particular f is strictly negative at sufficiently negative x . Since $f(0) > 0$ and f is continuous, the intermediate value theorem produces $c < 0$ with $f(c) = 0$, i.e. $I + cA$ is singular. □

Thank you

Questions?